

# DAY – 04 EXPERIMENTS

## 1. SUM OF N:

The screenshot shows a C++ IDE with three main panels. On the left, a sidebar contains 'QUESTIONS 21 / 48 completed' and a dropdown menu with 'Sum of N' selected. Below this, the title 'SUM OF N' is followed by the instruction: 'Write a program to print the numbers from 1 to 10 using a for loop.' The central 'PROGRAM EDITOR' panel shows the following C++ code:

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int i;
5     for(i=1;i<=10;i++){
6         cout<<i<<endl;
7     }
8 }
```

Below the editor is an 'OUTPUT' panel displaying the numbers 1 through 10, each on a new line. On the right, an 'INPUT' panel contains the text 'Your INPUT goes here!'.

## 2. TABLE:

The screenshot shows a C++ IDE with three main panels. On the left, a sidebar contains 'QUESTIONS 21 / 48 completed' and a dropdown menu with 'Table' selected. Below this, the title 'TABLE' is followed by the instruction: 'Write a program to read in an integer and print the multiplication table for that number using a for loop.' The central 'PROGRAM EDITOR' panel shows the following C++ code:

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n,i;
5     cin>>n;
6     for(i=1;i<=10;i++){
7         cout<<n<<" * "<<i<<" = "<<n*i<<endl;
8     }
9     return 0;
10 }
```

Below the editor is an 'OUTPUT' panel displaying the multiplication table for the number 7:

```
7 * 1 = 7
7 * 2 = 14
7 * 3 = 21
7 * 4 = 28
7 * 5 = 35
7 * 6 = 42
7 * 7 = 49
7 * 8 = 56
7 * 9 = 63
7 * 10 = 70
```

On the right, an 'INPUT' panel contains the number '7'.

### 3. SUM OF DIGITS:

QUESTIONS 21 / 48 completed

Sum of All

SUM OF ALL

Write a program to read in an integer and print the sum of all its digits using a while loop.

PROGRAM EDITOR

C++ Run Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n,digit,sum=0;
5     cin>>n;
6     while(n>0){
7         digit=n%10;
8         sum=sum+digit;
9         n=n/10;
10    }
11    cout<<"Sum of Digits = "<<sum;
12    return 0;
13 }
```

OUTPUT

Sum of Digits = 25

INPUT

71845

### 4. PALINDROME

QUESTIONS 21 / 48 completed

Palindrome

PALINDROME

Write a program to read in an integer and determine if it is a palindrome or not using a while loop.

PROGRAM EDITOR

C++ Run Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n,temp,reverse=0,rem;
5     cin>>n;
6     temp=n;
7     while(n!=0){
8         rem=n%10;
9         reverse=reverse*10+rem;
10        n=n/10;
11    }
12    if (temp==reverse)
13        cout<<temp<<" is a Palindrome";
14    else
15        cout<<temp<<" is not a Palindrome";
16    return 0;
17 }
```

OUTPUT

7 is a Palindrome

INPUT

7

## 5. GCD

QUESTIONS 21 / 48 completed

GCD

**GCD**

Write a program to read in two integers and compute their greatest common divisor using a do-while loop.

PROGRAM EDITOR

C++

Run Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int a,b,rem;
5     cin>>a;
6     cin>>b;
7     do{
8         rem=a%b;
9         a=b;
10        b=rem;
11    }
12    while(b!=0);
13    cout<<"GCD = "<< a;
14    return 0;
15 }
```

OUTPUT

GCD = 9

INPUT

18  
45

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